Appendix B: First-Principles Derivation of Constants (HMGD)

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Abstract

This paper proposes that the fundamental constants of physics—c, G, and h—are not arbitrary inputs but are emergent geometric and informational properties of the causal horizon. By defining the universe as a finite-information system bounded by the Hubble Radius (L_h), we derive the coupling strength of gravity (G), the quantization of action (h), and the relative weakness of gravitation (the Hierarchy Problem) from the discrete bit-density of the holographic screen. This provides a non-circular, bottom-up derivation of the physical parameters of our universe.

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1. The Axiomatic Substrate (δ and τ)

To avoid circular reasoning, we must establish the true foundational axioms of the universe's "hardware." In a discrete holographic substrate, we postulate the existence of a fundamental spatial resolution (δ, the distance between adjacent informational nodes) and a fundamental temporal update rate (τ, the discrete tick of the horizon's clock).

These two parameters are the *only* geometric priors in the framework. The speed of light c is not an axiom, but an emergent propagation limit defined by the substrate's resolution:

$$c = \frac{\delta}{\tau}$$

Therefore, c represents the global clock-speed of the informational horizon, derived purely from the underlying informational grid (δ, τ).

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2. Deriving Newton's Constant (G): The Area-Volume Coupling

In HMGD, gravity is an entropic force arising from the mapping of 3D volume information onto a 2D holographic boundary. Newton's constant G is the proportionality factor of this mapping.

According to the Equipartition Theorem, the total energy E of the system is distributed across the N informational bits on the horizon. Substituting the Unruh-Verlinde temperature ($T = \frac{\hbar a}{2\pi c k_B}$) and the holographic acceleration ($a_0 = c^2/2\pi L_h$), we identify that G emerges as the ratio between the global mass-energy and the surface information density, scaled by the **Holographic Fractal Index ($D_H = 2.5$)**:

$$G = \frac{c^3}{2\pi\hbar\Sigma \cdot (D_H/2)}$$

Where Σ is the informational flux (N/A) across the horizon. The inclusion of D_H accounts for the fractal distribution of information, bridging the gap between the 2D boundary and the 3D interior. G is therefore a geometric conversion factor between information (N) and energy (E), intrinsically linked to the quantum of action ($\hbar$).

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3. Deriving Planck's Constant (h): The Unit of Action

Planck's constant h defines the minimum unit of action ($Action = Energy \times Time$). In an informational universe, h is the action required to perform a single bit-flip (a state change) on the holographic horizon.

If the total action of the universe is the product of its total energy E_{tot} and its causal lifetime $t_h = L_h/c$, then h is defined rigorously as this total action divided by the total discrete informational capacity N of the causal horizon:

$$h = \frac{E_{tot} \cdot t_h}{N}$$

Crucially, this derivation does *not* rely on the Planck length (l_p). Instead, by equating N with the geometric area of the horizon divided by the fundamental pixel area δ^2, we find that the traditional "Planck length" $l_p = \sqrt{\frac{\hbar G}{c^3}}$ is merely an algebraic recombination of h, G, and c. The true fundamental length is the axiomatic δ. This identifies h as a **density-dependent action limit**, implying that quantum action is an emergent property of the "pixelated" nature of the causal boundary, free of circular dependency on l_p.

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4. The Fine-Structure Constant (α) and Electromagnetic Coupling

The fine-structure constant $\alpha \approx 1/137$ represents the coupling strength of the electromagnetic force. In the HMGD derivation, α is the **Informational Horizontal Coupling**. While gravity involves a projection *through* the volume, electromagnetism is an interaction *across* the 2D surface of the horizon.

α is derived as the ratio of the fundamental cross-section of a bit to the total informational width of the causal horizon. This explains why α is a dimensionless number; it is a purely geometric ratio of the "Pixel-to-Screen" interaction area.

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5. Resolving the Hierarchy Problem: Why Gravity is Weak

The "Hierarchy Problem" asks why the gravitational force is roughly 10^{40} times weaker than the electromagnetic force. HMGD resolves this by identifying the **Geometric Dilution** of the holographic projection.

While surface interactions (electromagnetism) scale with L_h, the gravitational projection involves the total bit-count N of the universe ($N \approx 10^{122}$). The "weakness" of gravity is a direct result of the energy of the universe being distributed across the massive informational capacity of the horizon:

$$\text{Coupling Ratio} \propto \frac{1}{\sqrt{N}} \approx 10^{-40}$$

Gravity is "weak" only because the informational horizon is "large." At the Planck scale ($N = 1$), gravity and electromagnetism achieve unification at a 1:1 coupling ratio.

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6. The Boltzmann Constant (k_B): The Energy-per-Bit

In HMGD, the Boltzmann constant k_B is not an arbitrary thermal factor, but the **Energy-per-Bit conversion factor**. It bridges the gap between informational entropy (S) and physical energy (E).

$$k_B = \frac{E_{bit}}{T_{horizon}}$$

This reveals that "Temperature" is simply the rate of informational flux, and k_B is the amount of work required to maintain one degree of freedom in the informational substrate.

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7. The Minimum Fermion Mass (Neutrinos)

The Standard Model fails to predict the mass of neutrinos, treating them initially as massless. In HMGD, the universal acceleration a_0 creates a non-zero "noise floor" for mass. Just as a wave bounded by L_h cannot have a frequency lower than c/L_h, a fermion coupled to the informational boundary cannot have a mass lower than the holographic metric bound.

The minimum mass coupling is derived geometrically from the quantum of action and the acceleration floor:

$$m_{min} \approx \sqrt{\frac{\hbar a_0}{c^3}}$$

Substituting the HMGD derivation for $a_0 = 1.04 \times 10^{-10}\ \mathrm{m/s^2}$:

$$m_{min} \approx \sqrt{\frac{(1.05 \times 10^{-34})(1.04 \times 10^{-10})}{(3 \times 10^8)^3}} \approx 6.3 \times 10^{-36} \text{ kg}$$

Converting to electron-volts ($E = mc^2$), this yields $m_\nu \approx 0.035$ eV. This result aligns with the observed cosmological bounds for the absolute neutrino mass scale, derived purely from the geometry of the Hubble radius without introducing a right-handed sterile neutrino.

8. The Elementary Charge (e) and the Alpha Coupling

In the Standard Model, the elementary charge e is an arbitrary input. In the HMGD framework, charge is not an intrinsic property of matter, but a **topological winding defect** within the informational grid of the causal horizon.

Crucially, the magnitude of e is not an independent constant but is strictly constrained by the global surface interaction ratio α. As derived in the companion microscopic framework (Paper II), α is a purely geometric result of the phase space equipartition of the Standard Model gauge groups:

$$\alpha = (4\pi^3 + \pi^2 + \pi)^{-1} \approx 1/137.036$$

The elementary charge e is thus recontextualized as the localized manifestation of this global topological coupling:

$$e^2 = 4\pi\epsilon_0\hbar c\alpha$$

By anchoring e to the geometric derivation of α, we remove the arbitrary nature of charge, proving it to be a structural artifact of the holographic screen's discrete phase space.

9. The Baryon-to-Photon Ratio (η)

The ΛCDM model offers no explanation for why there are roughly a billion photons for every baryon in the universe ($\eta \approx 6 \times 10^{-10}$). In HMGD, photons represent unentangled, 2D boundary information (the "surface" of the hologram), while baryons represent stable, complex 3D topological entanglements within the bulk volume.

The probability of the informational network forming a stable 3D bulk defect versus remaining a 2D surface excitation is governed by the Holographic Fractal Dimension ($D_H = 2.5$) and the topological winding ratio α. The baryon-to-photon ratio emerges strictly from the geometric probability limit of the substrate:

$$\eta \sim \mathcal{O}(\alpha^4) \approx 2.8 \times 10^{-9}$$

The precise ratio scales with the structural complexity required to bind quarks, proving that the sparsity of matter compared to radiation is a strict informational limit, not an arbitrary initial condition of the Big Bang.

10. Conclusion: The Unity of Physics

Fundamental constants c, G, h, α, k_B, m_ν, e, and η are emergent properties of a finite-information universe.

- c: The Logic Processing Speed.
- G: The Area-Volume Conversion (3D-to-2D).
- h: The Unit of Action (Bit-Flip).
- α: The Surface Interaction Ratio (Pixel-to-Screen).
- k_B: The Energy-per-Bit.
- m_ν: The Holographic Mass Floor.
- e: The Topological Winding Defect.
- η: The Bulk-to-Boundary Probability Limit.

By deriving these from the geometric constraints of the informational horizon L_h, HMGD establishes a deterministic, bottom-up foundation for all physical laws, removing the need for fine-tuning and providing a unified geometric framework for the dark sector and foundational physics.